

Decision Making with Sentiment Analysis in R

From Social Media Noise to Managerial Action

Mustafa Asım Ruhi

Data Analysis for Decision-Making

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Aim of the Presentation

Research question

How can customer tweets be transformed into decision-relevant insights for airline management?

Roadmap

1. Explain sentiment analysis
2. Show a reproducible R example
3. Translate results into managerial decisions
4. Let students try a short exercise

What is Sentiment Analysis?

Sentiment analysis is a text analysis method that identifies the emotional orientation of written language.

Typical categories:

- ▶ positive
- ▶ negative
- ▶ neutral

In business, it helps summarize customer opinions from reviews, tweets, surveys, and support messages.

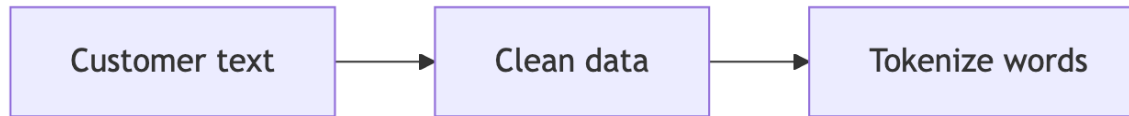
Why Does it Matter for Decision-Making?

Managers often face too much customer feedback to read manually.

Sentiment analysis helps to answer:

- ▶ Are customers mostly satisfied or dissatisfied?
- ▶ Which problems are repeated most often?
- ▶ Which operational area should management prioritize?

From Text to Decision



R Packages Used

```
library(tidyverse)
library(janitor)
library(tidytext)
library(lubridate)
library(scales)
```

The workflow mainly uses the tidyverse logic: import, clean, transform, visualize, and interpret.

Dataset

The example uses airline-related tweets.

airline_sentiment	n	share
negative	9178	62.7%
neutral	3099	21.2%
positive	2363	16.1%

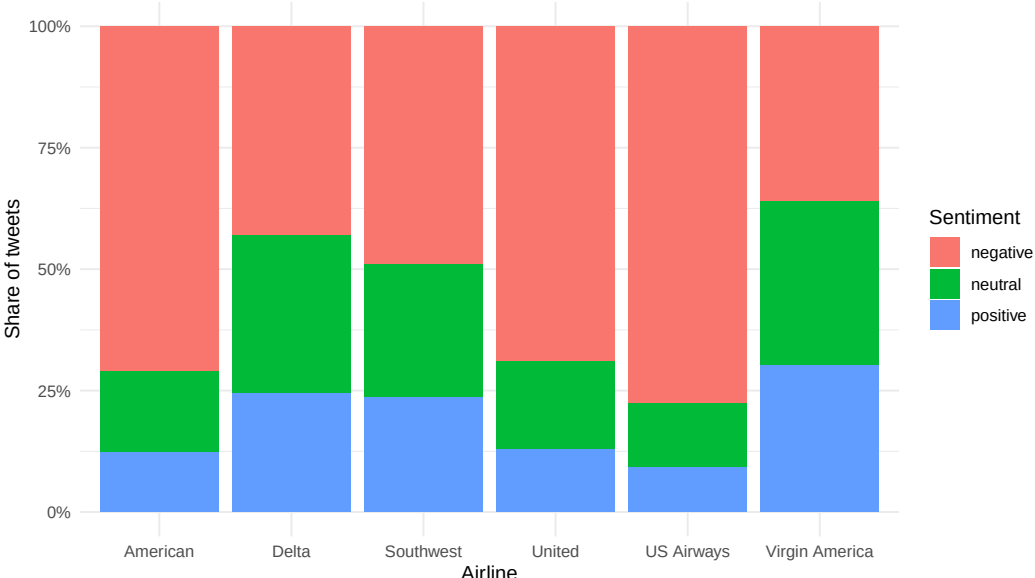
The dataset contains tweet text, airline names, sentiment labels, and reasons for negative tweets.

Step 1: Import and Clean Data

```
tweets <- readr::read_csv(url, show_col_types = FALSE) |>
  janitor::clean_names() |>
  select(
    airline_sentiment,
    negativereason,
    airline,
    retweet_count,
    text,
    tweet_created
  )
```

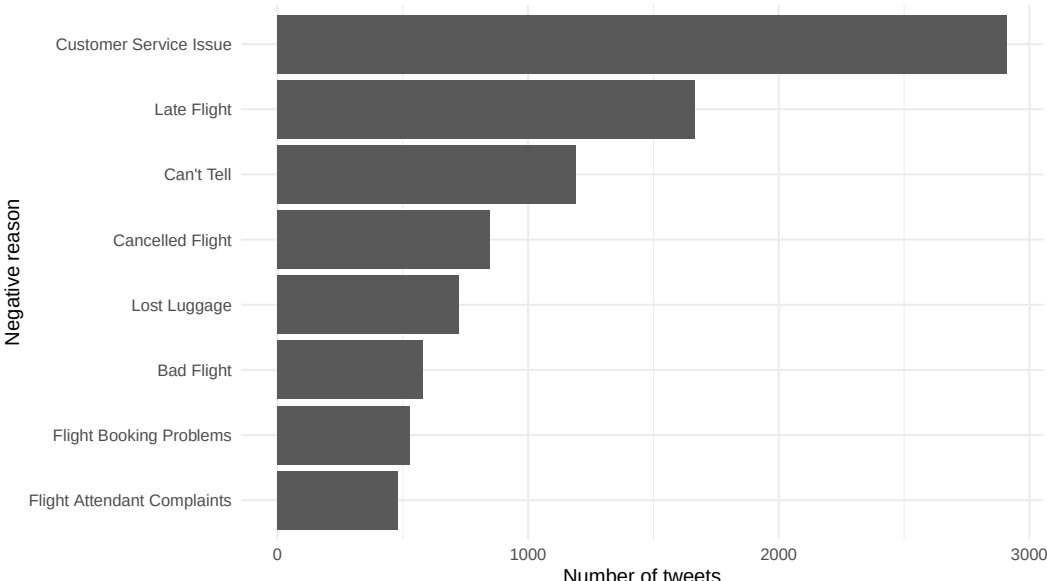

Step 2: Compare Sentiment by Airline

Customer sentiment differs across airlines



Step 3: Identify Negative Reasons

Main reasons behind negative customer sentiment



Interpretation

The analysis moves from general sentiment to concrete business problems.

Examples:

- ▶ Customer service issue -> support and service recovery
- ▶ Late flight -> punctuality and delay communication
- ▶ Lost luggage -> baggage tracking and recovery
- ▶ Cancelled flight -> rebooking and compensation

Step 4: Tokenize Tweet Text

```
tweets_words <- tweets |>
  mutate(tweet_id = row_number()) |>
  select(tweet_id, airline, airline_sentiment, text) |>
  unnest_tokens(word, text) |>
  anti_join(stop_words, by = "word")
```

Tokenization splits each tweet into individual words.

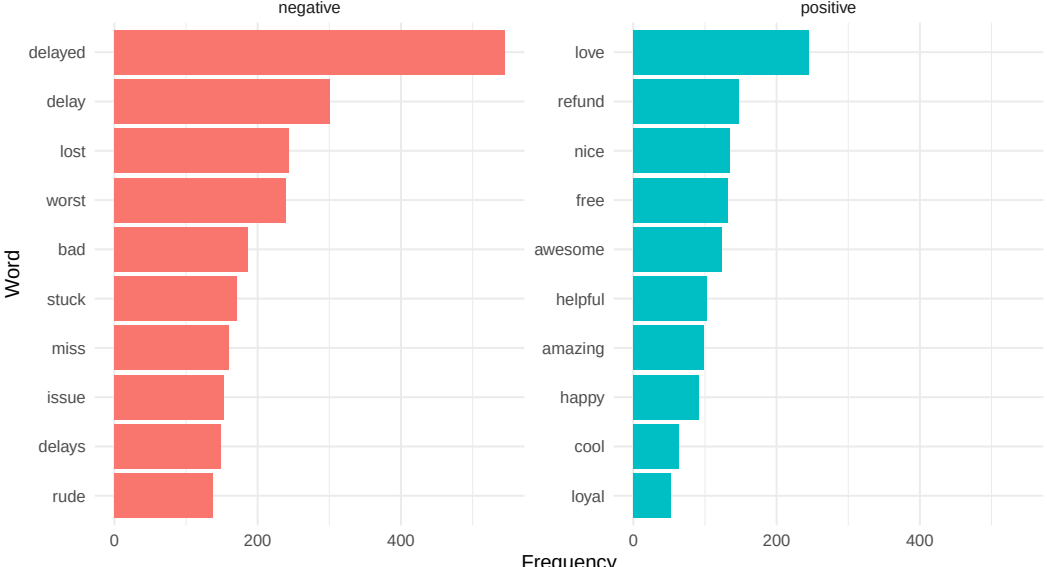
Step 5: Apply a Sentiment Lexicon

```
word_sentiments <- tweets_words |>  
  inner_join(get_sentiments("bing"), by = "word")
```

A lexicon-based approach matches words with a dictionary of positive and negative terms.

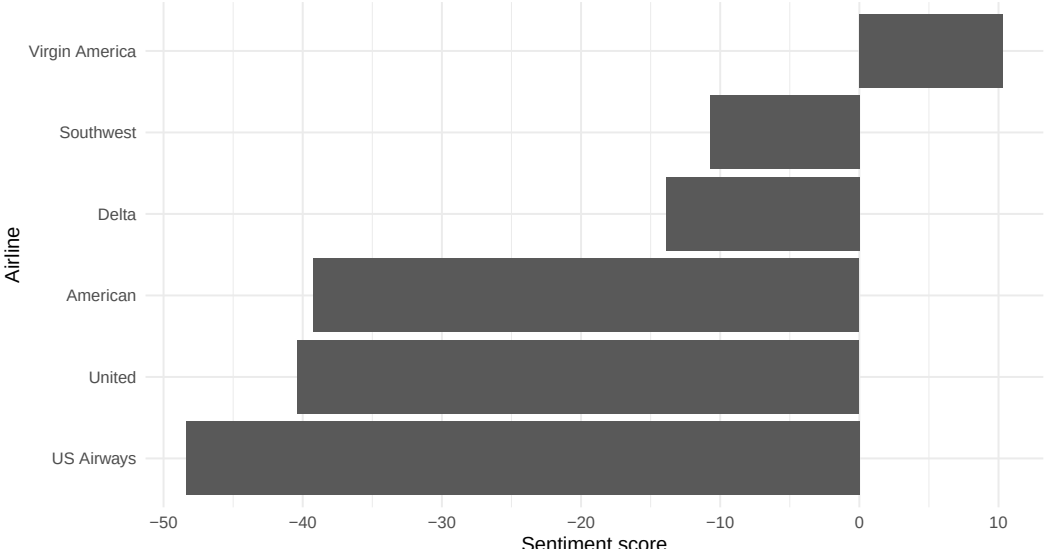
Most Frequent Sentiment Words

Most frequent positive and negative words



Sentiment Score by Airline

Lexicon-based sentiment score by airline
Positive words minus negative words per 100 sentiment words



From Analysis to Decision-Making

negativereason	n	managerial_action
Customer Service Issue	2910	Increase support capacity and service recovery.
Late Flight	1665	Improve punctuality monitoring and delay communication.
Can't Tell	1190	Investigate the complaint category further.
Cancelled Flight	847	Strengthen rebooking and compensation process.
Lost Luggage	724	Improve baggage tracking and recovery communication.
Bad Flight	580	Improve in-flight customer experience.

Key Decision Logic

The important question is not only:

What is the sentiment?

The better managerial question is:

What decision should this sentiment trigger?

Strengths

- ▶ Fast summary of large text data
- ▶ Reproducible workflow in R
- ▶ Useful for customer experience monitoring
- ▶ Helps prioritize operational problems

Limitations

- ▶ Sarcasm and irony are difficult
- ▶ Short tweets can be ambiguous
- ▶ Lexicons may miss context
- ▶ Human interpretation is still necessary

In-Class Exercise

Students will analyze three customer comments in R.

Task:

1. Tokenize comments
2. Remove stop words
3. Match words with a sentiment lexicon
4. Count positive and negative words
5. Recommend one managerial action

Exercise Code

```
library(tidyverse)
library(tidytext)

customer_comments <- tibble(
  text = c(
    "The flight was delayed and the service was terrible",
    "The staff was helpful and the flight was comfortable",
    "My luggage was lost and nobody helped me"
  )
)

customer_comments |>
  unnest_tokens(word, text) |>
  anti_join(stop_words, by = "word") |>
  inner_join(get_sentiments("bing"), by = "word") |>
  count(sentiment)
```

Conclusion

Sentiment analysis helps transform unstructured text into decision-relevant information.

The value is not the sentiment label itself.

The value is knowing which operational problem should be solved first.

References